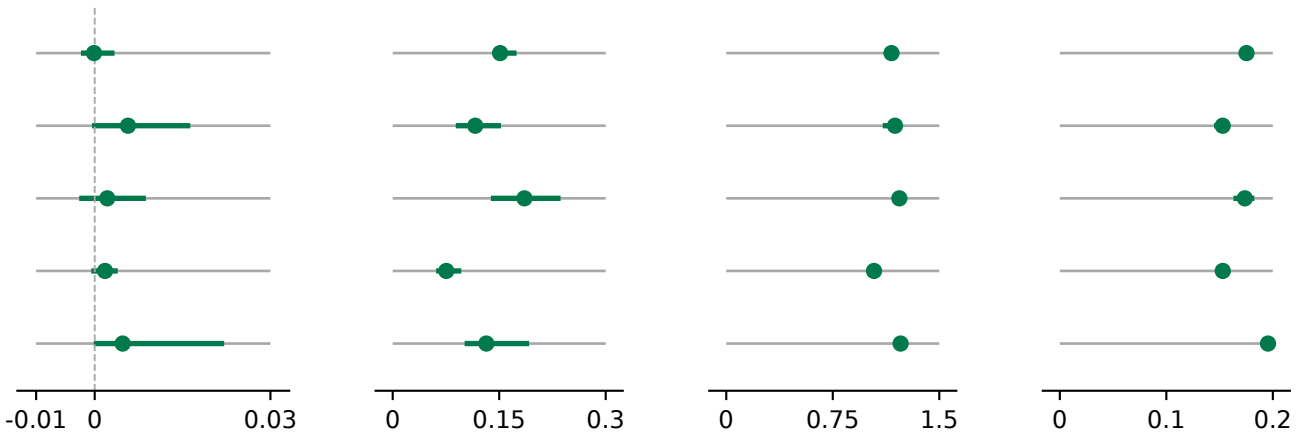
A RNA-seq configurations and fidelity

Cohort	Setup	Normalization
LAML	One-group VAE	raw
PAAD	One-group VAE	raw
READ	One-group VAE	raw
COAD	Two-group cVAE	TC
SKCM	Two-group cVAE	TC



Points: means; green lines: minimum–maximum across 5 subsampling draws.
ARI: zero = chance baseline. Lower MedAE is better. No AE-head augmentation.

B UMAP embeddings of empirical and generated RNA-seq samples

One-group VAE cohorts

LAML

VAE-1:100 · raw

PAAD

VAE-1:100 · raw

READ

VAE-1:100 · raw

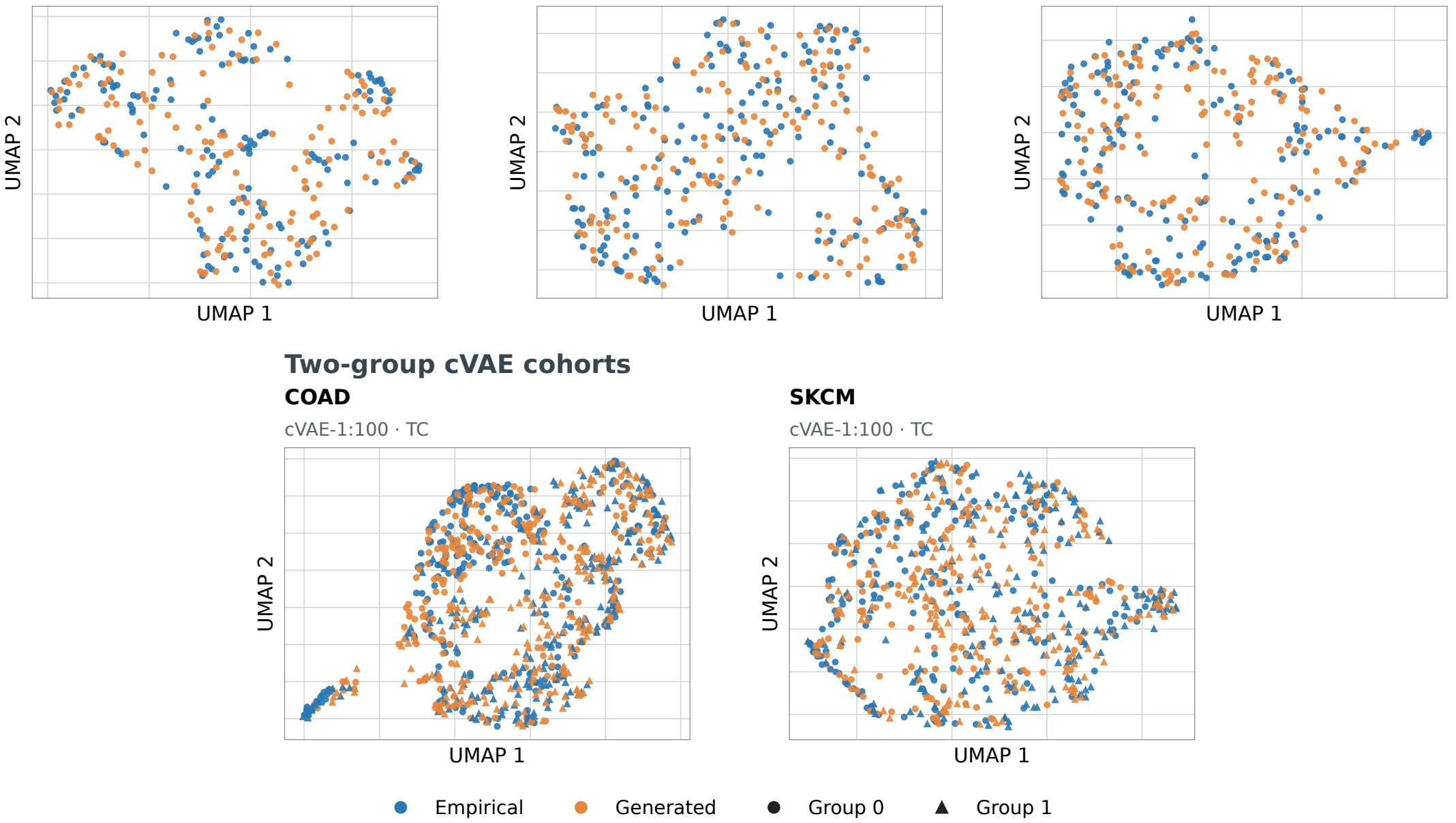
Two-group cVAE cohorts

COAD

cVAE-1:100 · TC

SKCM

cVAE-1:100 · TC



One-group samples are circles. Embeddings are fitted separately and are not comparable across cohorts.